

Applications of forces Mixed exercise 7

- 1 a Finding the components of P along each axis:

$$R(\rightarrow): P_x = 12 \cos 70^\circ + 10 \sin 75^\circ$$

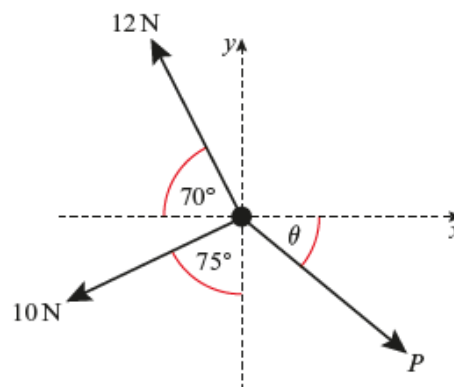
$$R(\uparrow): P_y = 12 \sin 70^\circ - 10 \cos 75^\circ$$

$$\tan \theta = \frac{P_y}{P_x}$$

$$\tan \theta = \frac{12 \sin 70^\circ - 10 \cos 75^\circ}{12 \cos 70^\circ + 10 \sin 75^\circ} = 0.63124...$$

$$\theta = 32.261...$$

The angle θ is 32.3° (to 3s.f.).



- b Using Pythagoras' theorem:

$$P^2 = P_x^2 + P_y^2$$

$$P^2 = (12 \cos 70^\circ + 10 \sin 75^\circ)^2 + (12 \sin 70^\circ - 10 \cos 75^\circ)^2$$

$$P = \sqrt{264.91...} = 16.276...$$

P has a magnitude of 16.3 N (3s.f.).

- 2 a $R(\nearrow)$:

$$W \cos \theta = 40 \sin 30^\circ + 30 \sin 45^\circ$$

$$R(\swarrow):$$

$$30 \cos 45^\circ + W \sin \theta = 40 \cos 30^\circ$$

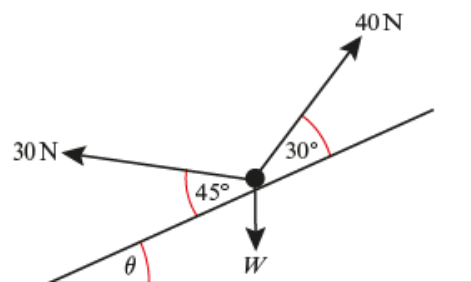
$$W \sin \theta = 40 \cos 30^\circ - 30 \cos 45^\circ$$

$$\frac{W \sin \theta}{W \cos \theta} = \frac{40 \cos 30^\circ - 30 \cos 45^\circ}{40 \sin 30^\circ + 30 \sin 45^\circ}$$

$$\tan \theta = \frac{20\sqrt{3} - 15\sqrt{2}}{20 + 15\sqrt{2}} = 0.35281...$$

$$\theta = 18.046...$$

The angle θ is 18.0° (to 3s.f.).



- b Using Pythagoras' theorem, $|W|^2$ is the sum of the squares of the two components.

$$W^2 = (20\sqrt{3} - 15\sqrt{2})^2 + (20 + 15\sqrt{2})^2$$

$$W = \sqrt{1878.8...} = 43.345...$$

The weight of the particle is 43.3 N (3s.f.).

3 Resolving horizontally:

$$T_1 \cos 20^\circ = T_2 \cos 10^\circ$$

$$T_2 = \frac{T_1 \cos 20^\circ}{\cos 10^\circ} \quad (1)$$

Resolving vertically:

$$T_1 \sin 20^\circ + T_2 \sin 10^\circ = 55g \quad (2)$$

Substituting $T_2 = \frac{T_1 \cos 20^\circ}{\cos 10^\circ}$ from (1) into (2):

$$T_1 \sin 20^\circ + \frac{T_1 \cos 20^\circ \times \sin 10^\circ}{\cos 10^\circ} = 55g$$

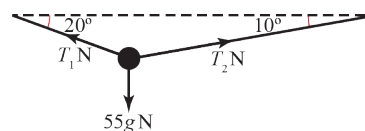
$$T_1 (\sin 20^\circ + \cos 20^\circ \tan 10^\circ) = 55g$$

$$T_1 = \frac{55 \times 9.8}{\sin 20^\circ + \cos 20^\circ \tan 10^\circ} = 1061.6...$$

Substituting this value of T_1 into (1):

$$T_2 = \frac{\cos 20^\circ \times 1061.6...}{\cos 10^\circ} = 1012.9...$$

The tensions in the two parts of the rope are 1062 N and 1013 N (nearest whole number).



4 Let the acceleration of the system be $a \text{ ms}^{-2}$

For the block, $R(\nearrow)$:

$$F = ma$$

$$T - 5g \sin 30^\circ = 5a$$

$$T = 5a + \frac{5g}{2} \quad (1)$$

For the pan and masses, $R(\downarrow)$:

$$F = ma$$

$$(2+5)g - T = (2+5)a$$

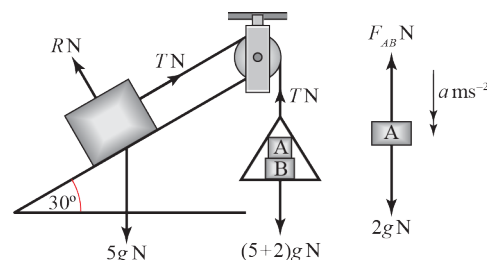
$$T = 7g - 7a \quad (2)$$

Since the string is inextensible, T is constant and hence (1) = (2):

$$5a + \frac{5g}{2} = 7g - 7a$$

$$12a = \frac{9g}{2}$$

$$a = \frac{3g}{8}$$



Hence, the scale-pan accelerates downward, away from the pulley with magnitude $a = \frac{3g}{8} \text{ ms}^{-2}$

Applying Newton's second law to mass A, and denoting the force exerted by B on A as F_{AB} :

$$2g - F_{AB} = 2a$$

$$F_{AB} = 2g - \frac{2 \times 3g}{8}$$

$$F_{AB} = \frac{10}{8}g$$

$$F_{AB} = 1.25 \times 9.8 = 12.25$$

- 4 However, Newton's third law of motion gives
 $|\text{force exerted by } B \text{ on } A| = |\text{force exerted by } A \text{ on } B|$.
 Therefore the force exerted by A on B is 12.25 N.

- 5 a For the 2 kg particle

$$\begin{aligned} R(\uparrow) \\ T - 2g &= 0 \\ \therefore T &= 2g \end{aligned}$$

For the 5 kg particle:

$$\begin{aligned} R(\nearrow) \\ T + F \cos \theta - 5g \sin \theta &= 0 \\ \therefore F \cos \theta &= 5g \sin \theta - T \end{aligned}$$

$$\text{As } T = 2g, \cos \theta = \frac{4}{5} \text{ and } \sin \theta = \frac{3}{5}$$

$$F \times \frac{4}{5} = 5g \times \frac{3}{5} - 2g$$

$$\therefore F = g \times \frac{5}{4}$$

$$= \frac{5g}{4}$$

$$= 12 \text{ (2 s.f.)}$$

- b $R(\nwarrow)$

$$R - F \sin \theta - 5g \cos \theta = 0$$

$$\therefore R = F \sin \theta + 5g \cos \theta$$

$$= \left(\frac{5}{4}g \times \frac{3}{5} \right) + \left(5g \times \frac{4}{5} \right)$$

$$= \frac{19}{4}g$$

$$= 47 \text{ (2 s.f.)}$$

- c F will be smaller

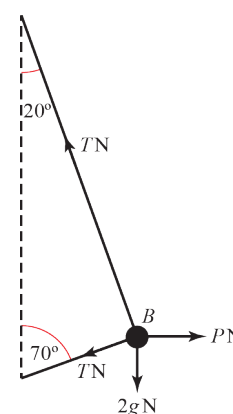
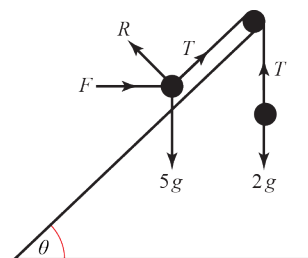
- 6 a Resolving vertically:

$$T \cos 20^\circ = T \cos 70^\circ + 2g$$

$$T(\cos 20^\circ - \cos 70^\circ) = 2g$$

$$T = \frac{2 \times 9.8}{\cos 20^\circ - \cos 70^\circ} = 32.793\dots$$

The tension in the string, to two significant figures, is 33 N.



- 6 b Resolving horizontally:

$$P = T \sin 20^\circ + T \sin 70^\circ$$

$$P = (\sin 20^\circ + \sin 70^\circ) \times 32.793...$$

$$P = 42.032...$$

The value of P is 42 N (2 s.f.).

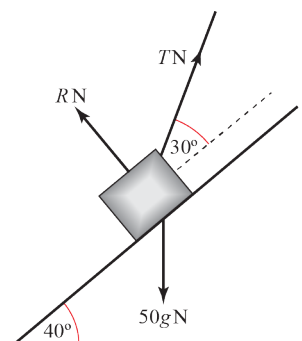
- 7 a $R(\nearrow)$:

$$T \cos 30^\circ = 50g \sin 40^\circ$$

$$T = \frac{50 \times 9.8 \sin 40^\circ}{\cos 30^\circ}$$

$$= 363.69...$$

The tension in the string is 364 N (3s.f.).



- b Even when the hill is covered in snow, there is likely to be some friction between the runners of the sled and the slope, so modelling the hill as a smooth slope is unrealistic.

- 8 Let S be the reaction of the wall on the ladder at B .
Let R be the reaction of the ground on the ladder at A .
(Both surfaces are smooth, so no friction.)

$$R(\rightarrow): F = S$$

Taking moments about A :

$$mg \times \frac{5}{2}a \times \cos \theta + F \times a \times \sin \theta = S \times 5a \times \sin \theta$$

$$\frac{5mg}{2} + F \tan \theta = 5S \tan \theta \quad (\text{dividing by } a \cos \theta)$$

$$\frac{5mg}{2} + F \tan \theta = 5F \tan \theta \quad (\text{Since } F = S)$$

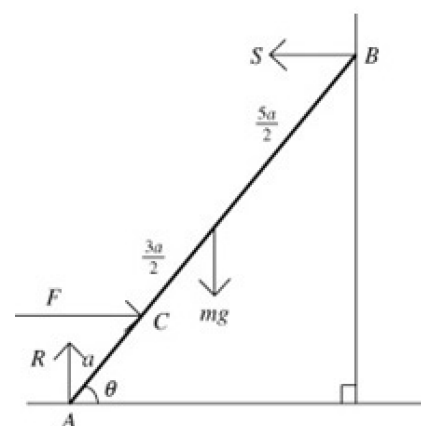
$$\frac{5mg}{2} = 4F \tan \theta$$

$$= 4 \times \frac{9}{5}F \quad (\text{Since } \tan \theta = \frac{9}{5})$$

$$= 7.2F$$

$$F = \frac{5mg}{2 \times 7.2}$$

$$= \frac{25mg}{72} \text{ as required.}$$



- 9 Let N be the reaction of the wall on the ladder at B .
 Let R be the reaction of the ground on the ladder at A ,
 Let F the friction between the ladder and the ground at A .

$$\tan \alpha = \frac{3}{4} \Rightarrow \sin \alpha = \frac{3}{5} \text{ and } \cos \alpha = \frac{4}{5}$$

$$R(\uparrow): R = mg + 2mg = 3mg$$

Taking moments about B :

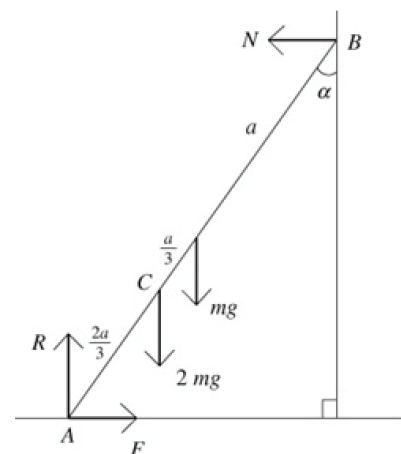
$$mg \times a \sin \alpha + 2mg \times \frac{4}{3}a \sin \alpha + F \times 2a \cos \alpha = R \times 2a \sin \alpha$$

$$mga \times \frac{3}{5} + \frac{8mga}{3} \times \frac{3}{5} + F \times 2a \times \frac{4}{5} = 6mga \times \frac{3}{5}$$

$$F \times \frac{8a}{5} = \frac{18mga}{5} - \frac{8mga}{5} - \frac{3mga}{5}$$

$$F \times \frac{8a}{5} = \frac{7mga}{5}$$

$$F = \frac{7mga}{5} \times \frac{5}{8a} \\ = \frac{7mg}{8}$$



The ladder and the child are in equilibrium, so

$$F \leq \mu R$$

$$\frac{7mg}{8} \leq \mu \times 3mg$$

$$\mu \geq \frac{7}{24}$$

The least possible value for μ is $\frac{7}{24}$

- 10 Let R be the reaction of the ground on the ladder at A ,
 Let N be the reaction of the wall on the ladder at B .
 Let F be the friction between the wall and the ladder at B .

- a Since you do not know the magnitude of F , you cannot resolve vertically to find R .

Therefore, take moments about B (since this eliminates F):

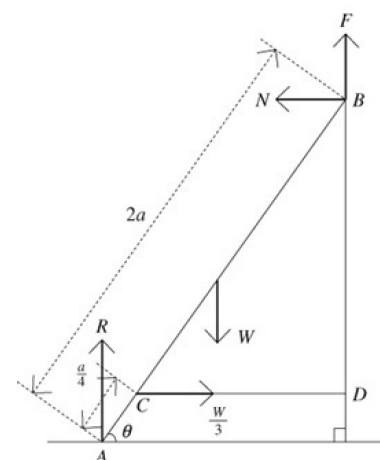
$$\frac{W}{3} \times \frac{7a}{4} \sin \theta + W \times a \cos \theta = R \times 2a \cos \theta$$

$$\frac{7W}{12} \times \tan \theta + W = 2R \quad (\text{dividing through by } a \cos \theta)$$

$$\frac{7}{12} \times \frac{4}{3} W + W = 2R \quad (\text{since } \tan \theta = \frac{4}{3})$$

$$\frac{16W}{9} = 2R$$

$$R = \frac{8W}{9}$$



10 b $R(\rightarrow): N = \frac{W}{3}$

$R(\uparrow):$

$R + F = W$

$F = W - R$

$= W - \frac{8}{9}W$

$= \frac{W}{9}$

For the ladder to remain in equilibrium,

$F \leq \mu N$

$\frac{W}{9} \leq \mu \frac{W}{3}$

$\mu \geq \frac{1}{3}$

c The ladder had negligible thickness / the ladder does not bend.

11 a Let S be the reaction of the wall on the ladder

Let R be the reaction of the ground on the ladder

Let F the friction between the ladder and the ground

Let X be the point where the lines of action of R and S intersect, as shown in the diagram.

By Pythagoras's Theorem, distance from base of ladder to wall is 3 m.

$R(\rightarrow): F = S$

$R(\uparrow): R = W$

Taking moments about X :

$1.5W = 4F$

Suppose the ladder can rest in equilibrium in this position. Then

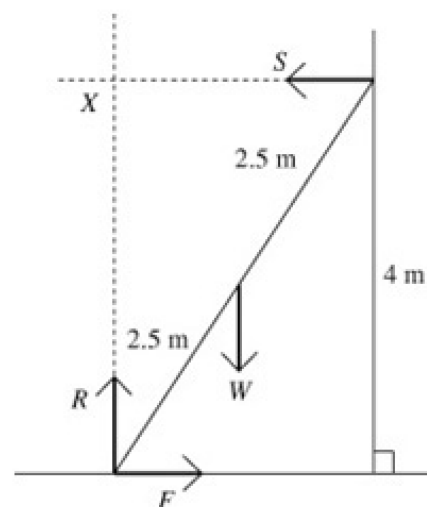
$F \leq \mu R$

$\frac{1.5W}{4} \leq 0.3 \times W$

$\frac{3W}{8} \leq \frac{3W}{10}$

$30 \leq 24$

which is false, therefore the assumption that $F \leq \mu R$ must be false – the ladder cannot be resting in equilibrium.



- 11 b** With the brick in place, take moments about X :

$$1.5W = 4F \text{ so}$$

$$F = \frac{1.5W}{4} = \frac{3W}{8}$$

which is independent of M , the mass of the brick.

c $R(\uparrow) \quad R = W + Mg$

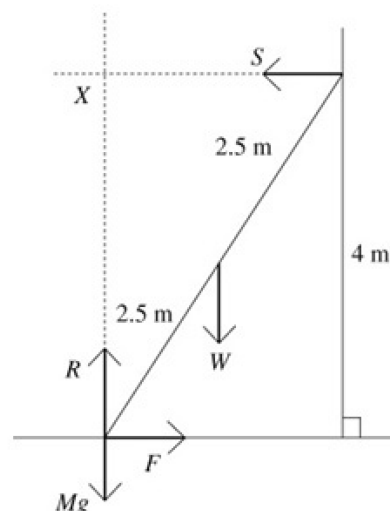
$$R(\rightarrow) \quad F = S$$

$$F \leq \mu R \Rightarrow \frac{3W}{8} \leq 0.3(W + Mg) = \frac{3(W + Mg)}{10}$$

$$\Rightarrow 10W \leq 8W + 8Mg$$

$$8Mg \geq 2W, \quad M \geq \frac{W}{4g}$$

So the smallest value for M is $\frac{W}{4g}$



- 12** Let S be the reaction of the wall on the ladder at Q
Let R be the reaction of the ground on the ladder at P

$$\tan \alpha = \frac{5}{2} \Rightarrow \sin \alpha = \frac{5}{\sqrt{29}} \text{ and } \cos \alpha = \frac{2}{\sqrt{29}}$$

Since the ladder is in limiting equilibrium,
frictional force at the wall = $\mu S = 0.2S$.

Taking moments about P :

$$20g \times 1 \cos \alpha = S \times 4 \sin \alpha + 0.2S \times 4 \times \cos \alpha$$

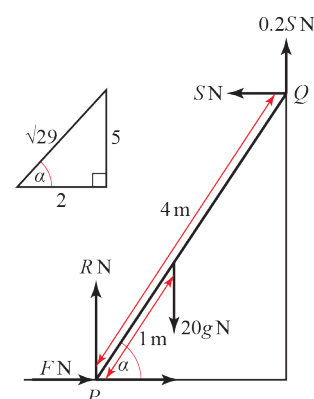
$$\frac{20 \times 2}{\sqrt{29}} g = \left(\frac{4 \times 5}{\sqrt{29}} + \frac{0.8 \times 2}{\sqrt{29}} \right) S$$

$$40g = 21.6S$$

$$S = \frac{392}{21.6} = 18.148...$$

$$R(\rightarrow): F = S$$

The force F required to hold the ladder still is 18 N (2 s.f.).



- 13** Since the rod is uniform, the weight acts from the midpoint of AB .

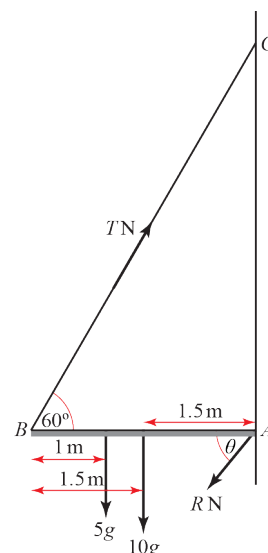
- a** Take moments about A :

$$(10g \times 1.5) + (5g \times 2) = T \times 3 \sin 60^\circ$$

$$T = \frac{25g}{3 \sin 60^\circ}$$

$$T = \frac{25 \times 9.8 \times 2}{3\sqrt{3}} = 94.300...$$

The tension in the string is 94.3 N (3s.f.).



- b** Let the horizontal and vertical components of the reaction R at the hinge be R_x and R_y respectively.

Resolving horizontally:

$$R_x = T \cos 60^\circ$$

$$R_x = \frac{25g}{3 \sin 60^\circ} \cos 60^\circ$$

$$R_x = \frac{25g}{3 \tan 60^\circ} = \frac{25g}{3\sqrt{3}}$$

Resolving vertically upwards:

$$R_y = T \sin 60^\circ - 10g - 5g$$

$$R_y = \frac{25g}{3 \sin 60^\circ} \sin 60^\circ - 15g$$

$$R_y = \left(\frac{25}{3} - 15 \right) g = -\frac{20g}{3}$$

The reaction at the hinge is given by:

$$R^2 = R_x^2 + R_y^2$$

$$R^2 = \left(\frac{25g}{3\sqrt{3}} \right)^2 + \left(-\frac{20g}{3} \right)^2$$

$$R^2 = \left(\frac{625}{27} + \frac{400}{9} \right) g^2$$

$$R = \sqrt{\frac{1825}{27}} \times 9.8 = 80.570...$$

$$\tan \theta = \frac{R_y}{R_x}$$

$$\tan \theta = \frac{20g}{3} \times \frac{3\sqrt{3}}{25g} = \frac{4\sqrt{3}}{5}$$

$$\theta = 54.182...$$

The reaction at the hinge is 80.6 N acting at 54.2° below the horizontal (both values to 3 s.f.).

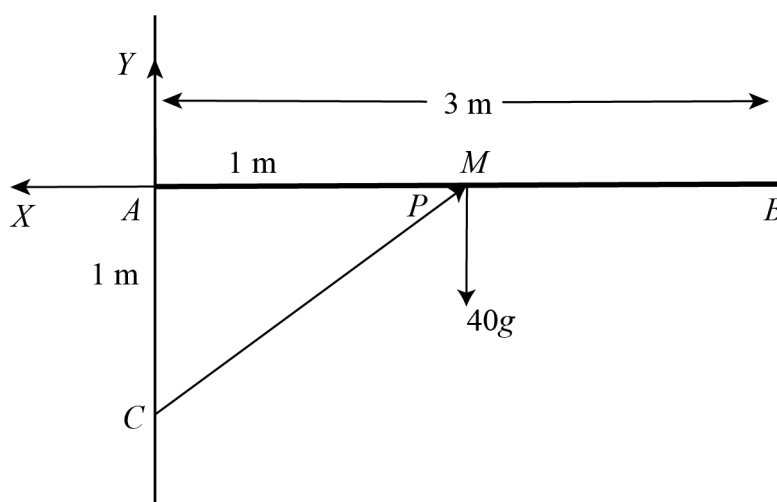
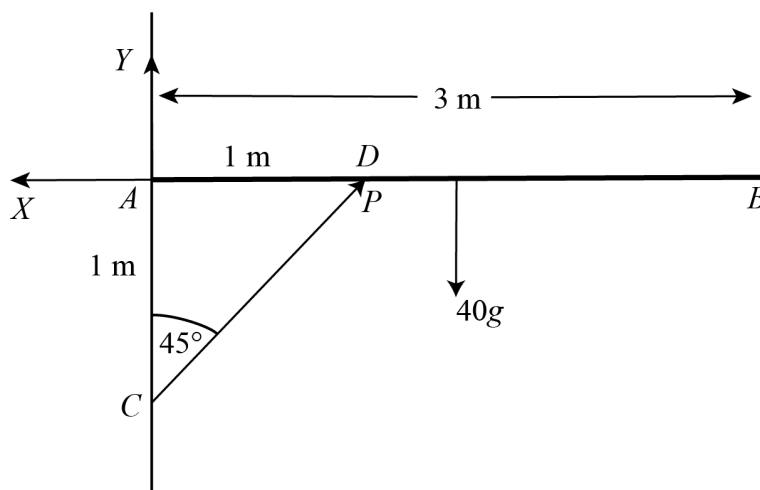
- 14 Let the horizontal and vertical components of the force at A be X and Y respectively. Let the thrust in the rod be P .

a $M(A): 1 \times P \times \cos 45^\circ = 40g \times \frac{3}{2}$
 $P = \frac{60g}{\cos 45^\circ} = 60\sqrt{2}g = 830 \text{ N (2 s.f.)}$

b $R(\rightarrow): X = P \cos 45^\circ = 60g$
 $R(\uparrow): Y + P \cos 45^\circ = 40g$
 $Y = 40g - 60g = -20g$

resultant $= \sqrt{X^2 + Y^2}$
 $= 10g\sqrt{4^2 + 2^2} = 10g\sqrt{40}$
 $= 620 \text{ N (2 s.f.)}$

- c The lines of action of P and the weight meet at M , hence the line of action of the resultant of X and Y must also pass through M (3 forces acting on a body in equilibrium). Therefore the reaction must act horizontally (i.e. no vertical component).



$$15 \quad \tan \alpha = \frac{3}{4} \Rightarrow \sin \alpha = \frac{3}{5} \quad \text{and} \quad \cos \alpha = \frac{4}{5}$$

$$u = 0 \text{ ms}^{-1}, s = 6 \text{ m}, t = 1.5 \text{ s}, a = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$6 = 0 + \frac{1}{2}a \times 1.5^2 = \frac{9a}{8}$$

$$a = 6 \times \frac{8}{9} = \frac{16}{3}$$

$$R(\nearrow): R = 3g \cos \alpha$$

$$R(\swarrow):$$

$$F = ma$$

$$3g \sin \alpha - \mu R = 3 \times \frac{16}{3}$$

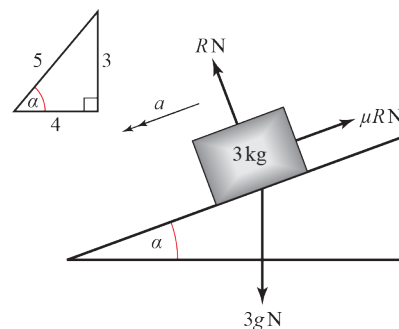
$$3g \sin \alpha - (\mu \times 3g \cos \alpha) = 16$$

$$\frac{3}{5}g - \frac{4}{5}\mu g = \frac{16}{3}$$

$$9g - 12\mu g = 80$$

$$\mu = \frac{(9 \times 9.8) - 80}{12 \times 9.8} = 0.06972...$$

The coefficient of friction is 0.070 (3s.f.).



$$16 \quad R(\nearrow): R = 5g \cos 30^\circ + 80 \sin 10^\circ$$

$$R(\swarrow):$$

$$ma = F$$

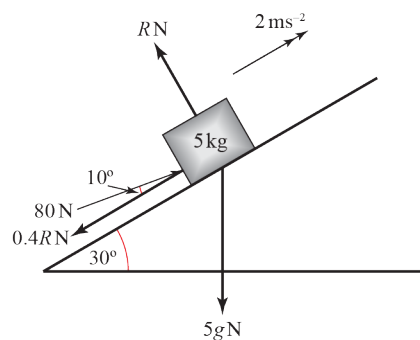
$$5a = 80 \cos 10^\circ - 0.4R - 5g \sin 30^\circ$$

$$5a = 80 \cos 10^\circ - 0.4(5g \cos 30^\circ + 80 \sin 10^\circ) - 5g \sin 30^\circ$$

$$a = 16 \cos 10^\circ - 0.4(g \cos 30^\circ + 16 \sin 10^\circ) - g \sin 30^\circ$$

$$a = 6.3507...$$

The acceleration of the particle is 6.53 ms^{-2} (3s.f.).



17 Since $m_2 > \mu m_1$, when system is released from rest then B moves downwards and A moves towards the pulley P

For particle A :

$$R(\uparrow): R = \mu m_1 g$$

$$R(\rightarrow):$$

$$F = m_1 a$$

$$T - \mu R = m_1 a$$

$$T - \mu m_1 g = m_1 a$$

$$T = m_1 a + \mu m_1 g \quad (1)$$

For particle B :

$$R(\uparrow):$$

$$F = m_2 a$$

$$m_2 g - T = m_2 a$$

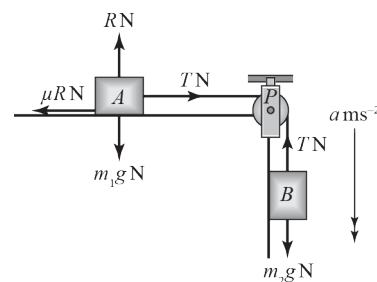
$$T = m_2 g - m_2 a \quad (2)$$

Since string is inextensible, the values of T are identical, so $(1) = (2)$:

$$m_1 a + \mu m_1 g = m_2 g - m_2 a$$

$$m_1 a + m_2 a = m_2 g - \mu m_1 g$$

$$a = \frac{g(m_2 - \mu m_1)}{m_1 + m_2} \quad \text{as required.}$$



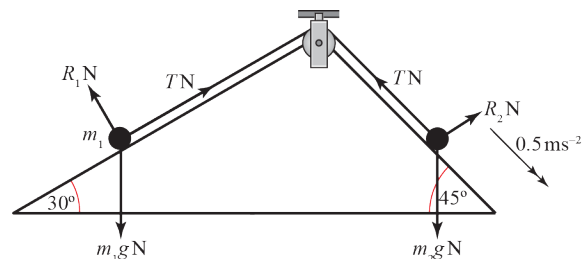
18 For particle with mass m_1 :

$R(\nearrow)$

$$F = ma$$

$$T - m_1 g \sin 30^\circ = \frac{m_1}{2}$$

$$T = \left(\frac{1}{2} + g \sin 30^\circ \right) m_1 \quad (1)$$



For particle with mass m_2 :

$R(\nearrow)$

$$F = ma$$

$$m_2 g \sin 45^\circ - T = \frac{m_2}{2}$$

$$\left(g \sin 45^\circ - \frac{1}{2} \right) m_2 = T \quad (2)$$

Since string is inextensible, T is constant throughout and hence (1) = (2):

$$\left(\frac{1}{2} + g \sin 30^\circ \right) m_1 = \left(g \sin 45^\circ - \frac{1}{2} \right) m_2$$

$$\left(\frac{1}{2} + \frac{g}{2} \right) m_1 = \left(\frac{g\sqrt{2}}{2} - \frac{1}{2} \right) m_2$$

$$\frac{m_1}{m_2} = \frac{g\sqrt{2} - 1}{1 + g} \text{ as required}$$